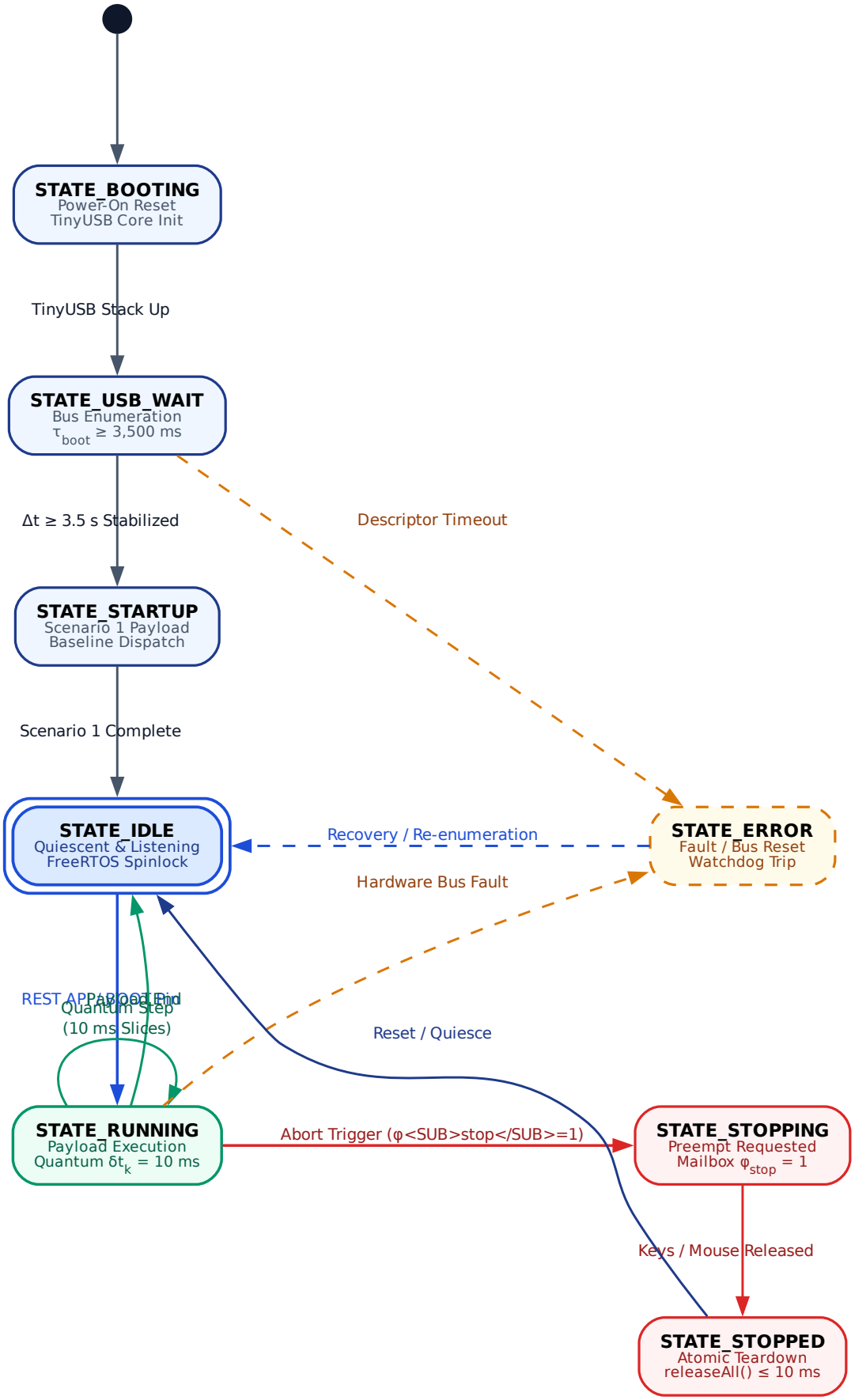
STATE_BOOTING
Power-On Reset
TinyUSB Core Init
TinyUSB Stack Up
STATE_USB_WAIT
Bus Enumeration
τboot ≥ 3,500 ms
Δt ≥ 3.5 s Stabilized
Descriptor Timeout
STATE_STARTUP
Scenario 1 Payload
Baseline Dispatch
Scenario 1 Complete
STATE_IDLE
Quiescent & Listening
FreeRTOS Spinlock
Recovery / Re-enumeration
STATE_ERROR
Fault / Bus Reset
Watchdog Trip
Hardware Bus Fault
REST APPROACH
Quantum Step
(10 ms Slices)
Reset / Quiesce
STATE_RUNNING
Payload Execution
Quantum δtk = 10 ms
Abort Trigger (φ_{stop}=1)
STATE_STOPPING
Preempt Requested
Mailbox φstop = 1
Keys / Mouse Released
STATE_STOPPED
Atomic Teardown
releaseAll() ≤ 10 ms